

Road-Accident Risk Map Squares

A human-readable guide to **cells_features_v1.parquet**

The file has **416,428** rows. Each row is one small square of the map of the United States (each square is about 0.005 degrees on a side, roughly half a kilometre). Instead of listing individual accidents, the file counts how many accidents happened in each square in 2017-2020, describes what those accidents were like (night, bad weather, near a junction, and so on), and records how many happened afterwards in 2021-2022. A square is marked **high-risk** if it had more than 16 accidents in 2021-2022.

What this PDF contains: (1) plain-English meaning of every column, (2) key numbers, (3) three charts, (4) a random sample of squares written out in words and percentages, (5) the 25 highest-risk squares, and (6) the squares where history looked quiet but the next two years were dangerous.

What it does not contain: all 416,428 rows. Printed in full they would fill roughly 8,000 pages, so this PDF shows samples and summaries. The complete table can be given as a CSV or Excel file instead.

1. What each column means

Column in the file	Plain-English meaning
iy, ix	The square's position on the map grid (row and column number). Together they are a unique ID for the square.
lat, lng	Latitude and longitude of the centre of the square.
n_hist	Number of accidents in this square from 2017 to 2020 (all four years combined).
active_years	How many of those four years had at least one accident here (1 to 4).
n_per_year	Average accidents per year in 2017-2020 (just n_hist divided by 4).
neighbor_n	Accidents in the 8 squares touching this one (2017-2020). A rough measure of how busy the surrounding area is.
sev34_share	Share of this square's accidents that were severe (severity level 3 or 4).
night_share	Share of accidents that happened at night.
bad_weather_share	Share of accidents that happened in bad weather.
Junction / Crossing / Traffic_Signal / Stop / Railway / Give_Way / Roundabout / Bump / Traffic_Calming_share	Share of accidents that happened next to that road feature: a junction, pedestrian crossing, traffic signal, stop sign, railway crossing, give-way sign, roundabout, speed bump, or traffic-calming measure.
target_n	Number of accidents in this square in 2021-2022. This is what the model tries to anticipate.
high_risk	1 (shown as YES) if target_n is above 16, otherwise 0 (no). This is the answer the model learns to predict.
confidence	How much past history the square has: Low, Medium or High. A low-confidence square has few past accidents, so its prediction is less reliable.
low_history	1 if confidence is Low, otherwise 0. It carries the same information as confidence.
split	Whether the square is used for training the model (train) or for testing it afterwards (test).
demo_low_history_high_risk	A flag for squares that had very little history (low confidence) yet turned out to be high-risk. It is worked out from the answer, so it should NOT be used as a model input.

Share columns are fractions from 0 to 1 in the file (for example 0.5 means half of that square's accidents); in this PDF they are shown as percentages.

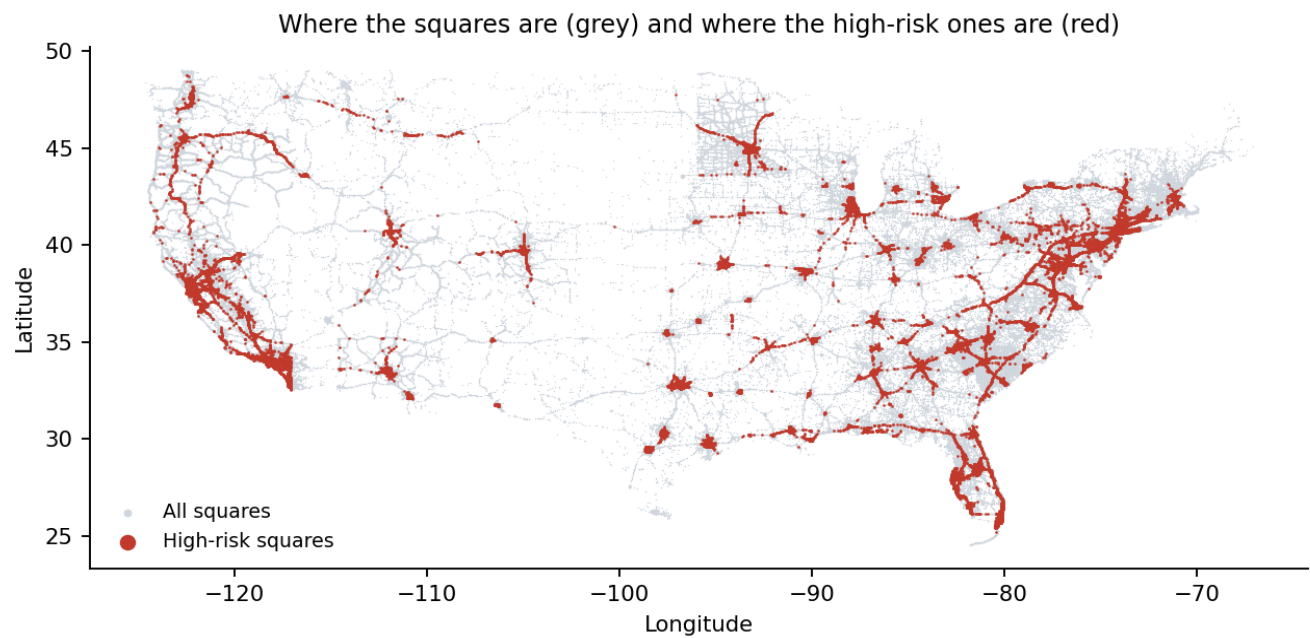
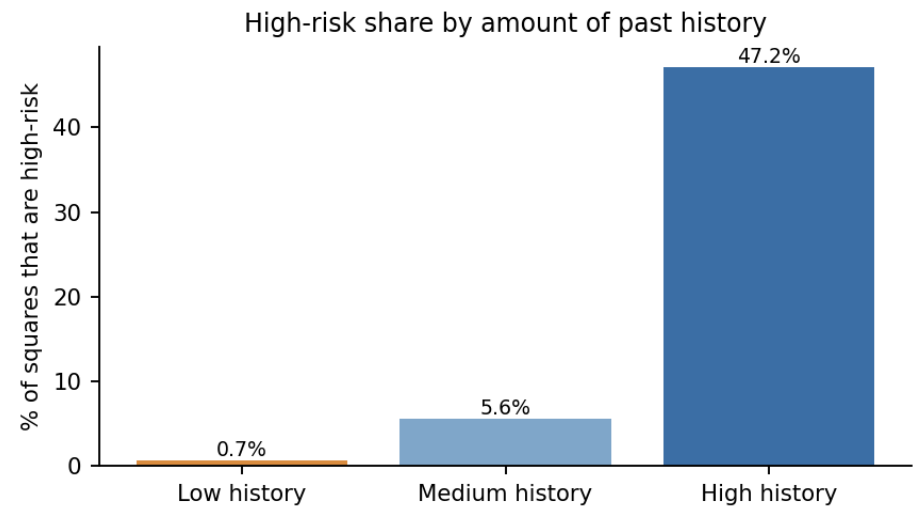
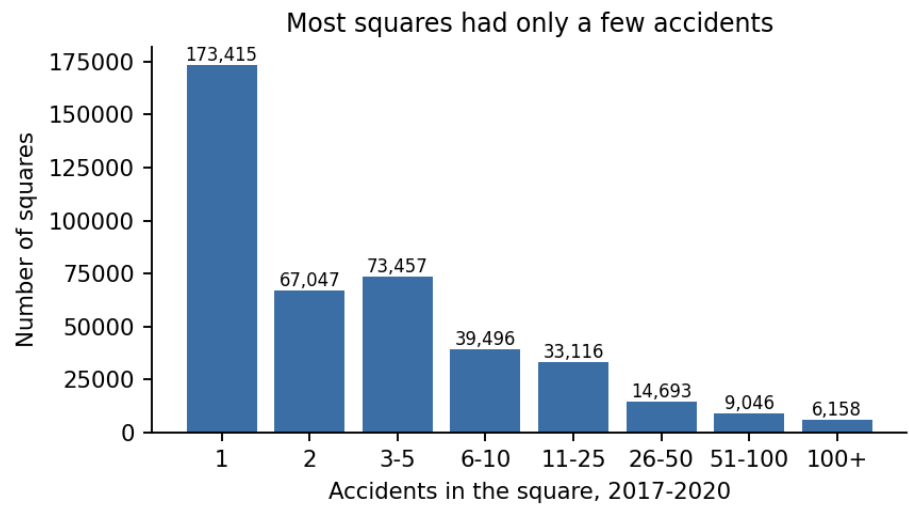
2. Key numbers

Measure	Value
Squares (rows)	416,428
Accidents counted in 2017-2020 (sum of n_hist)	3,744,735
Accidents counted in 2021-2022 (sum of target_n)	2,896,449
Typical square, 2017-2020 (median / average / maximum)	2 / 9.0 / 1,565
Typical square, 2021-2022 (median / average / maximum)	1 / 7.0 / 1,921
High-risk squares (more than 16 accidents in 2021-2022)	39,888 (9.6% of squares)
Training squares / test squares	332,731 / 83,697
High-risk share in training / test	9.6% / 9.7%
Low / Medium / High history squares	240,462 / 107,696 / 68,270
Squares flagged as low-history but high-risk	249
Latitude range	24.56 to 49.00
Longitude range	-124.62 to -67.11
Missing values	None

Average share of accidents by condition (across all squares)

Condition	Average share	Condition	Average share
Severe (sev. 3-4)	16.8%	Near stop sign	4.7%
Night	33.4%	Near railway	0.8%
Bad weather	11.6%	Near give-way sign	0.4%
Near junction	2.6%	Near roundabout	0.0%
Near crossing	10.4%	Near speed bump	0.1%
Near traffic signal	12.5%	Near traffic calming	0.1%

3. Charts



4. A random sample of 25 squares, in plain form

Random sample (fixed seed, so it is repeatable). Percentages are the share of that square's 2017-2020 accidents.

Square ID (row, col)	Location (lat, lng)	Count 2017-20	Years with accidents (of 4)	Nearby count*	Severe (sev. 3-4)	At night	Bad weather	Near junction	Near signal	Count 2021-22	High risk?	History level	Set
6584, -16116	32.9225, -80.5775	1	1	13	0%	0%	0%	0%	0%	1	no	Low	train
8215, -18517	41.0775, -92.5825	1	1	0	0%	0%	0%	0%	0%	0	no	Low	train
7643, -24529	38.2175, -122.6425	2	2	1	0%	100%	0%	0%	0%	6	no	Low	train
6897, -16032	34.4875, -80.1575	1	1	0	0%	0%	0%	0%	0%	0	no	Low	train
6278, -18957	31.3925, -94.7825	1	1	0	0%	0%	0%	0%	0%	0	no	Low	test
5669, -16438	28.3475, -82.1875	2	1	6	0%	100%	0%	0%	0%	2	no	Low	train
8610, -15232	43.0525, -76.1575	57	4	174	68%	14%	16%	0%	11%	15	no	High	train
5441, -16280	27.2075, -81.3975	5	2	3	0%	80%	0%	0%	0%	3	no	Medium	train
7121, -19518	35.6075, -97.5875	36	4	160	8%	11%	8%	0%	22%	9	no	High	train
6976, -16530	34.8825, -82.6475	9	4	0	0%	22%	33%	0%	0%	6	no	Medium	train
8443, -17658	42.2175, -88.2875	3	2	0	67%	33%	0%	0%	67%	0	no	Medium	train
8059, -15201	40.2975, -76.0025	2	1	5	0%	100%	0%	100%	0%	11	no	Low	test
8319, -15460	41.5975, -77.2975	4	1	3	0%	0%	0%	0%	0%	0	no	Medium	test
6820, -23640	34.1025, -118.1975	239	4	146	5%	33%	8%	33%	0%	77	YES	High	train
7938, -20977	39.6925, -104.8825	1	1	39	0%	0%	0%	0%	0%	0	no	Low	train
5351, -16361	26.7575, -81.8025	2	1	1	0%	50%	0%	0%	0%	1	no	Low	train
7530, -24206	37.6525, -121.0275	9	3	43	0%	67%	22%	0%	0%	27	YES	Medium	train
8424, -23955	42.1225, -119.7725	1	1	1	0%	100%	100%	0%	0%	0	no	Low	train
7721, -24304	38.6075, -121.5175	3	2	198	0%	0%	0%	0%	0%	0	no	Medium	test
7993, -21037	39.9675, -105.1825	1	1	25	0%	0%	100%	0%	0%	1	no	Low	test
8078, -22384	40.3925, -111.9175	4	2	13	0%	25%	0%	0%	50%	6	no	Medium	train
7994, -24441	39.9725, -122.2025	1	1	0	0%	100%	0%	0%	0%	0	no	Low	train
7818, -19153	39.0925, -95.7625	1	1	0	0%	0%	0%	0%	0%	0	no	Low	train
9033, -24787	45.1675, -123.9325	4	2	4	0%	75%	0%	0%	0%	4	no	Medium	train
7242, -17766	36.2125, -88.8275	1	1	0	0%	0%	0%	0%	0%	0	no	Low	train

Count = number of accidents in the square. *Nearby count = accidents in the 8 surrounding squares, 2017-2020.

5. The 25 highest-risk squares (most accidents in 2021-2022)

Square ID (row, col)	Location (lat, lng)	Count 2017-20	Years with accidents (of 4)	Nearby count*	Severe (sev. 3-4)	At night	Bad weather	Near junction	Near signal	Count 2021-22	High risk?	History level	Set
5177, -16033	25.8875, -80.1625	115	4	221	1%	30%	2%	0%	13%	1,921	YES	High	train
5689, -16295	28.4475, -81.4725	97	4	487	24%	47%	9%	5%	44%	1,563	YES	High	train
5178, -16038	25.8925, -80.1875	118	4	377	0%	25%	5%	0%	19%	1,513	YES	High	train
5690, -16281	28.4525, -81.4025	111	4	124	5%	18%	6%	0%	41%	1,398	YES	High	train
5684, -16281	28.4225, -81.4025	223	4	361	0%	12%	4%	0%	34%	1,367	YES	High	train
5137, -16084	25.6875, -80.4175	435	4	283	0%	12%	7%	0%	33%	1,266	YES	High	train
5149, -16068	25.7475, -80.3375	197	4	92	1%	13%	1%	0%	39%	1,217	YES	High	train
5188, -16062	25.9425, -80.3075	188	4	80	1%	15%	9%	0%	43%	1,194	YES	High	train
5176, -16049	25.8825, -80.2425	114	4	90	0%	4%	6%	0%	32%	1,191	YES	High	train
5146, -16068	25.7325, -80.3375	253	4	218	0%	13%	6%	0%	23%	1,120	YES	High	train
5137, -16074	25.6875, -80.3675	136	4	81	1%	18%	4%	0%	37%	1,116	YES	High	train
5690, -16296	28.4525, -81.4775	129	4	504	1%	18%	12%	0%	59%	1,048	YES	High	train
5153, -16074	25.7675, -80.3675	119	4	219	0%	27%	10%	0%	39%	1,023	YES	High	train
5690, -16295	28.4525, -81.4725	120	4	464	68%	39%	8%	0%	6%	1,021	YES	High	train
5713, -16242	28.5675, -81.2075	204	4	88	1%	17%	4%	0%	44%	1,018	YES	High	train
5189, -16030	25.9475, -80.1475	209	4	118	0%	11%	5%	0%	30%	1,018	YES	High	train
5140, -16067	25.7025, -80.3325	269	4	230	0%	6%	15%	0%	9%	1,003	YES	High	train
5713, -16258	28.5675, -81.2875	180	4	85	1%	20%	4%	0%	49%	900	YES	High	train
5137, -16077	25.6875, -80.3825	283	4	551	1%	12%	14%	0%	58%	898	YES	High	train
5139, -16077	25.6975, -80.3825	173	3	221	0%	5%	5%	0%	5%	886	YES	High	train
5689, -16296	28.4475, -81.4775	192	4	438	1%	2%	4%	0%	14%	856	YES	High	train
5152, -16061	25.7625, -80.3025	33	3	36	3%	30%	6%	0%	18%	828	YES	High	train
5185, -16031	25.9275, -80.1525	140	4	165	1%	15%	4%	0%	27%	821	YES	High	train
5136, -16084	25.6825, -80.4175	177	3	546	0%	2%	10%	0%	0%	819	YES	High	train
5178, -16033	25.8925, -80.1625	153	3	233	1%	12%	3%	0%	8%	814	YES	High	train

6. Quiet history, dangerous future: 25 of the 249 flagged squares

These squares had only 1 or 2 accidents in 2017-2020 but more than 16 in 2021-2022. They are the hardest cases for any model, because the past gave almost no warning. Shown here: the 25 with the most accidents in 2021-2022.

Square ID (row, col)	Location (lat, lng)	Count 2017-20	Years with accidents (of 4)	Nearby count*	Severe (sev. 3-4)	At night	Bad weather	Near junction	Near signal	Count 2021-22	High risk?	History level	Set
5671, -16299	28.3575, -81.4925	1	1	127	0%	0%	0%	0%	0%	492	YES	Low	train
5677, -16301	28.3875, -81.5025	2	1	147	0%	0%	0%	0%	0%	300	YES	Low	train
5308, -16368	26.5425, -81.8375	2	1	191	0%	50%	0%	0%	0%	162	YES	Low	train
5124, -16077	25.6225, -80.3825	1	1	537	0%	0%	0%	0%	0%	154	YES	Low	train
5571, -16465	27.8575, -82.3225	2	1	201	0%	0%	0%	0%	0%	142	YES	Low	train
6804, -23655	34.0225, -118.2725	2	1	988	0%	50%	0%	0%	0%	115	YES	Low	train
5155, -16054	25.7775, -80.2675	1	1	209	0%	0%	0%	0%	0%	105	YES	Low	train
6802, -23685	34.0125, -118.4225	2	1	757	0%	50%	0%	0%	0%	104	YES	Low	train
5309, -16372	26.5475, -81.8575	1	1	138	0%	0%	0%	0%	0%	98	YES	Low	train
6808, -23664	34.0425, -118.3175	2	2	500	0%	50%	0%	0%	50%	97	YES	Low	train
6554, -23415	32.7725, -117.0725	1	1	284	0%	0%	0%	0%	100%	96	YES	Low	train
6830, -23693	34.1525, -118.4625	2	2	1,724	0%	50%	0%	0%	0%	89	YES	Low	train
5579, -16468	27.8975, -82.3375	1	1	202	0%	0%	0%	0%	0%	88	YES	Low	train
5152, -16077	25.7625, -80.3825	2	2	312	0%	50%	0%	0%	0%	82	YES	Low	train
5187, -16062	25.9375, -80.3075	2	1	280	0%	0%	0%	0%	0%	80	YES	Low	train
5617, -16490	28.0875, -82.4475	2	1	347	0%	0%	0%	0%	0%	76	YES	Low	train
8197, -14733	40.9875, -73.6625	1	1	270	0%	0%	0%	0%	0%	70	YES	Low	train
5267, -16362	26.3375, -81.8075	2	2	171	0%	0%	0%	0%	0%	69	YES	Low	train
5181, -16079	25.9075, -80.3925	2	1	130	0%	50%	0%	0%	0%	69	YES	Low	train
5142, -16057	25.7125, -80.2825	2	1	165	0%	50%	0%	0%	0%	68	YES	Low	train
7384, -23960	36.9225, -119.7975	2	2	121	0%	0%	50%	0%	0%	67	YES	Low	train
6790, -23511	33.9525, -117.5525	1	1	136	0%	0%	0%	0%	0%	62	YES	Low	train
5489, -16508	27.4475, -82.5375	1	1	157	0%	0%	0%	0%	100%	60	YES	Low	train
6583, -23425	32.9175, -117.1225	2	1	317	0%	0%	0%	0%	100%	59	YES	Low	test
8129, -14775	40.6475, -73.8725	2	2	234	50%	0%	0%	0%	0%	58	YES	Low	train